DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #05D: NON-WEAPON EQUIPMENT & PROCEDURAL SYSTEM (PART 4 — MULTI-PIECE SET & SYNERGY ARCHITECTURE: 62 RESONANCE SETS, PAIRED ITEMS, CURSED/SOUL-BOUND GEAR, SANCTUM CATALYSTS 1–3★ & PRIMORDIAL TRINITY)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Master Roadmap (`05A–05G`)**:
  - `05A`: Group 1 — Body Armor & Protective Attire
  - `05B`: Group 2 — Personal Jewelry & Wearables + 4 Legendary Accessories
  - `05C`: Group 3 — Wearable Talismans, Portable Formation Cores & Profession/Exploration Tools
  - **`05D` (This Document)**: Group 4 — **Multi-Piece Equipment Set & Synergy Systems (`62` Resonance Sets `RES_*` & `RES_NINE_DAO_*`, Paired Items `E03`, Cursed/Soul-Bound `E04`, Sanctum Catalyst Star Attunement `1–3★`, & 3/3 Legendary Primordial Trinity)**
  - `05E`: Part 5 — Holistic Overview of All Non-Weapon Equipment
  - `05F`: Part 6 — Penultimate Deep Audit: Procedural Equipment Engine & Templates
  - `05G`: Part 7 — Grand Finale: Master End-to-End Audit of the Entire Equipment System

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **Save/Load Stat Invariance (`Zero Post-Load Stat Drift`)**: An item refined to `3★` (`q = 1.85`) MUST have the **exact same `it.st` values before and after `SIM.serialize` $\to$ `SIM.deserialize`**!

3. **Preserve `resonance_tactics_assembly_suite.js` & `v320` Contracts**: All 62 Resonance Sets (19 baseline + 43 Pagoda `RES_NINE_DAO_*`) covering all 286 subpaths must continue passing their 2-piece and 3-piece assertions.

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1. [SET-RES-01] Four Logic & Reachability Bugs in the 62 Equipment Resonance Sets (`M.itemMatchesResonanceSet` & `NINE_DAO_PAGODA`)

1.1 Clashing-Element Items Still Activate Elemental Sets (`js/martial.js:323–334`)!

- **Exact Code (`martial.js:330–333`)**:

```javascript
const elKey = it.elementalRole === 'EXPLICIT_NON_ELEMENTAL' ?
'EXPLICIT_NON_ELEMENTAL' : (it.element || 'NEUTRAL');
if (Array.isArray(rs.elementFamily) && (rs.elementFamily.includes(elKey) || ...))
return true;
if (n && n.cult && Array.isArray(rs.subpaths) && rs.subpaths.includes(n.cult.sub))
return true;
```

- **Logical Bug**: Line 332 returns `true` whenever `rs.subpaths.includes(n.cult.sub)`, **completely bypassing the `rs.elementFamily` check**! An `ELIXIR_DAO` (`FIRE`) alchemist wearing a random `WATER_ICE` Robe, a `YIN_SHADOW` Focus, and a `BLOOD_DEMON` Ring (three clashing elements with no `resonanceSetId`!) still gets **full 3/3 Flame-Cauldron Set activation**!
- **Fix**: Only allow the `rs.subpaths` fallback on items without an explicit `resonanceSetId` when `elKey === 'NEUTRAL'` or `rs.elementFamily` is empty—otherwise require element compatibility (preserving Pagoda pieces because `pagodaSetPieces` stamps explicit `resonanceSetId: 'RES_NINE_DAO_' + pathId`, which returns `true` at line 327!).

1.2 Missing `it.legendary` Universal Set Wildcard (`combat-design` Sec 3.4 vs `martial.js:323–334`)

- Project documentation specifies: *"Universal Wildcard: Any equipped legendary gear (`it.legendary === true`) counts as a valid match for ANY of the Equipment Resonance Sets"*.

- Yet `M.itemMatchesResonanceSet` (`lines 323–334`) **never checks `it.legendary`**! Upgrading 1 piece of a 3/3 Resonance Set to an off-type/off-element Legendary Relic **breaks the 3/3 Resonance Set**! Add `if (it.legendary) return true;` right after the `minTier` check at line 326.

1.3 `34` of the `43` Pagoda Resonance Sets (`RES_NINE_DAO_<PATH>`) Never Drop in Live Play (`js/explore.js:742`)!

- `MARTIAL.RESONANCE_SETS` registers all `43` `RES_NINE_DAO_<PATH>` sets (`Floors 1–43`), **but `EX.tryVolunteer` (`explore.js:742`) hardcodes `{ maxFloor: 9 }`** (`02B`)! Autonomous parties never climb Floors `10–43`, leaving **79% (`34/43`) of Pagoda Resonance Sets unobtainable**! Scale `{ maxFloor }` dynamically up to `43` based on party stage/power.

1.4 Unscaled 3-Piece Set Rider DoT & Cache Durability Key

- Scale `bonus3.rider` DoT potency (`0.012`) by `totalScale` (`1 + (avgTier - 3) * 0.12`) so Tier-7 3-piece sets deal stronger elemental riders than Tier-3 sets, and include `usableMask` in `_resSetCache` / `_legResCache` so mid-fight durability breaks immediately drop set bonuses even before `_passVer` increments.

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2. `[SET-PAIR-02]` Asymmetric Element-Order Bug in Paired Items (`E03`, `js/economy.js:1113–1124`) & Zero Static `pairId` Twins!

- **Exact Code (`js/economy.js:1119`)**:

```javascript
if (w && x && ((w.subpath && w.subpath === x.subpath) || (validEl(w.element) &&
validEl(x.element) && (
  (w.element === 'LIGHT_YANG' && (x.element === 'DARK_YIN' || x.element ===
'YIN_SHADOW')) ||
  (w.element === 'FIRE' && x.element === 'WATER_ICE') ||
  w.element === x.element
))))
```

- **Bug 2A — Non-Commutative Element Check**:
  - `WEAPON = 'FIRE'` + `ACCESSORY = 'WATER_ICE'` $\to$ **`active: true` (`bonusMult: 1.06, +3 crit, +4 pen`)**!

    - **`WEAPON = 'WATER_ICE'` + `ACCESSORY = 'FIRE'` $\to$ `active: false` (`0%` bonus!)**!
    - `WEAPON = 'LIGHT_YANG'` + `ACCESSORY = 'YIN_SHADOW'` $\to$ **`active: true`**!
    - **`WEAPON = 'YIN_SHADOW'` + `ACCESSORY = 'LIGHT_YANG'` $\to$ `active: false` (`0%` bonus!)**!
    - **Fix**: Make Yin-Yang (`LIGHT_YANG` $\leftrightarrow$ `YIN_SHADOW`/`DARK_YIN`) and Fire-Ice (`FIRE` $\leftrightarrow$ `WATER_ICE`) symmetric in **both** directions!
- **Bug 2B — Zero Authored `pairId` Twins in the Real Catalog**:
    - `w.pairId === x.pairId` (`1.16×` ศาสตราคู่ประสานหยินหยาง) and `w.pairId === a.pairId` (`1.15×` ศาสตราเกราะคู่ประสาน) exist in code, **yet `0` real catalog items carry `pairId`!**
    - Assign canonical `pairId`s to iconic lore pairs (e.g., `LEG_WPN_BODHI_NIRVANA_ROSARY` + `LEG_ACC_BODHI_STARLIGHT_BEADS` `PAIR_BODHI_108`; `LEG_WPN_JIUXIAO_PHOENIX_ZITHER` + `LEG_ARM_CELESTIAL_CRANE_ROBE` `PAIR_NINE_HEAVENS_MELODY`; `LEG_WPN_CHUNQIU_JUDGMENT_BRUSH` + `LEG_ACC_TAICHI_HEAVEN_SEAL` `PAIR_IMPERIAL_MANDATE`; `LEG_WPN_OATHBREAKER_DRAGON_LANCE` + `LEG_ACC_DRAGON_PHOENIX_BANGLE` `PAIR_DRAGON_COVENANT`).

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3. `[SET-CAT-03]` Save/Load Stat-Deletion Bug on `3★` Catalyst-Refined Legendary Relics (`economy.js:522–540` vs `visitors_clues.js:682–695`)!

Trace what happens when a Legendary Relic (`def.q = 1.35`) is refined with `CAT_PRIMORDIAL_FLAME_6` up to **`3★` (`q = 1.85`)**:

1. **Before Save/Load (`economy.js:522–540`)**:
    - `ECO.refineGearWithCatalyst` scales `gear.st[k]` directly by `ratio = 1.85 / 1.35 = 1.370×` (`e.g. Yitian Sword atk: 680 -> 931.8`), while `gear._vcNorm` (`non-enumerable!`) is `1` so `ST.gearStats` (`visitors_clues.js:684`) immediately returns `gear.st` (`931.8`).
2. **After Save & Reload (`SIM.serialize` $\to$ `SIM.deserialize`, `visitors_clues.js:652, 684–694`)**:
    - Because `_vcNorm` is `enumerable: false`, `LoadedGear._vcNorm` is `undefined`!
    - On the first `ST.gearStats(LoadedGear)` call, line 684 (`if (it.st && it._vcNorm)`) is **`false`**, so execution enters line 687:

```javascript
const qRatio = clamp(VC.gearQuality(it) / (def.q || 1.35), 0.75, 1.25); // <-- HARD-CLAMPED AT 1.25!!
if (it.st && it._stQ === qRatio) return it.st;
```

  - Because `it._stQ` was `1.0` (`!== 1.25`), `ST.gearStats` **overwrites `it.st` using `1.25×` (`850 ATK`) instead of `1.37×` (`931.8 ATK`), silently deleting `-81.8 ATK` simply because the player saved and reloaded the game**!
- **Exact Fix**:
  1. In `visitors_clues.js:687`, raise the upper `qRatio` clamp from `1.25` to `1.40` (`1.85 / 1.35 = 1.3704`) so `3★` (`q = 1.85`) is fully supported!
  2. In `ECO.refineGearWithCatalyst` (`economy.js:535`), when refining a `VC.LEGENDARY_WEAPON_BY_ID[gear.id]` item, recompute `gear.st` cleanly via `delete gear.st; delete gear._vcNorm; ST.gearStats(gear);` so pre-save and post-load stats are **100% identical to the tenth decimal place**!
  3. Require `FORGING >= 35` (or `forge` officer assistance) in `ECO.findCatalystRefinementCandidate` (`economy.js:563` & `cognition.js:705`) so unskilled novices don't burn Tier-6 Primordial Catalysts!

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4. `[SET-CURSE-04]` Dormant Cursed & Soul-Bound Lifecycle (`E04`, `economy.js:1126–1130`) & Quadruple Set-Stacking Guardrails

1. **Complete the Cursed / Soul-Bound Loop (`E04`)**:
   - Currently `0` items spawn with `cursed: true`, `drawbackHpDrainPct` (`0.03` / `0.05`) never drains HP in combat (`equip_repro_result.log:11`), and no purification rite sets `cursePurified = true`!
   - Allow Dark Sanctum / `IMMORTAL_TOMB` / `NETHER_GRAVE` drops (`12%` chance on `BLOOD_DEMON`/`YIN_SHADOW` T4+ loot) to roll `cursed: true` (`+28%` power, `-3% MaxHP` per combat round in `C.fightStep` + `+0.8 heartDemon` unless purified at `shrine` by a `BUDDHA_DHARMA`/`SAGE_SUPPORT` elder!), and penalize unauthorized wielders of `soulBoundOwnerId` items in `M.gearSuitability` (`score - 500`)!
2. **Soft-Cap Simultaneous 4-Layer Set Stacking**:
   - Because `3/3 Resonance Set` + `3/3 Primordial Trinity` + `Named Paired Items (1.16×)` + `Subpath Equip Bonus` can all fire simultaneously on the same 3 equipment slots (`+60%..+80%` compounded multipliers!), apply a clean diminishing-returns curve when both `legRes.active`

and `resSet.active` are active (`secondary set bonus * 0.65`) so progression remains rewarding without 1-shot inflation.

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5. Implementation & Verification Checklist for GPT / Gemini (`05D`)

- [ ] **`M.itemMatchesResonanceSet` (`martial.js:323–334`)**: Add `if (it.legendary) return true;` wildcard; prevent clashing-element non-set items from matching via `rs.subpaths`; unlock Pagoda Floors `10–43` in `explore.js:742`; scale 3-piece `rider` DoT with `tierScale`.
- [ ] **`ECO.evaluatePairedItems` (`economy.js:1119`)**: Fix symmetric `WATER_ICE` $\leftrightarrow$ `FIRE` and `YIN_SHADOW` $\leftrightarrow$ `LIGHT_YANG` order check, and add canonical `pairId`s to lore-matched Legendary Relic pairs.
- [ ] **`ECO.refineGearWithCatalyst` (`economy.js:522` & `visitors_clues.js:687`)**: Raise `qRatio` ceiling to `1.40` and unify `ST.gearStats` rebuild so `3★` (`q = 1.85`) stats never drop on Save/Load; gate autonomous refinement on `FORGING` skill.
- [ ] **Cursed / Soul-Bound (`E04`) & Stacking Cap**: Apply `drawbackHpDrainPct` in combat rounds, add `shrine` curse-purification & `M.gearSuitability` curse/soulbound awareness, and soft-cap overlapping `Trinity + Resonance Set` stacking.